

TATA CLIQ – DATA ARCHITECTURE & DAX STRATEGY DOCUMENT

E-Commerce Sales, Customer & Product Analytics | Power BI

Purpose: provide a handover-ready architecture reference covering source, tables, facts/dimensions, columns, data types, relationships, schema, ER view, Power BI connectivity, references, and a DAX implementation strategy.

1. Source & Evidence

Item	Observed / documented value	Status
Report	TATA CLIQ Analytics Dashboard	Confirmed
PBIX model tables	FactSales, DimDate, DimCustomer, DimProduct, DimEmployee, DimStore, FactReturns, SalesTargets, RegionTarget, ViewSelector	Confirmed from model layout
Business scope	Sales, profitability, customers, products, returns and delivery	Confirmed
Original physical source	Not exposed in the supplied user guide; PBIX connection metadata does not identify a source such as SQL/Excel in the extracted connection artifact	Needs validation
Storage mode	Not asserted from supplied evidence	Needs validation

The supplied Power BI user guide states that the report tracks sales, profitability, customers and product performance and lists the model tables above.

2. Data Model – Fact vs Dimension

Table	Role	Business purpose
FactSales	FACT	Core sales transactions; sales, profit, orders, returns/delivery-related metrics
FactReturns	FACT	Return transaction/detail data
DimDate	DIMENSION	Calendar attributes: year, month, day
DimCustomer	DIMENSION	Customer, region, state, city
DimProduct	DIMENSION	Product, category, sub-category, brand
DimEmployee	DIMENSION	Employee, department, region
DimStore	DIMENSION	Store and state
RegionTarget	TARGET / DIM-LIKE	Regional target sales and profit
SalesTargets	TARGET	Target values used for target analysis; exact physical relationship requires validation
ViewSelector	HELPER / DISCONNECTED	Likely used for view-selection logic; validate physical relationships

The guide explicitly describes FactSales as the core transaction table and identifies the five main dimensions plus target tables.

3. Column Inventory

Table	Columns / fields identified	Data type classification	Usage
DimDate	Year, Month, Day	Integer / Text / Date (validate exact model types)	Calendar filtering
DimCustomer	CustomerID, CustomerName, Region, State, City	Text / Integer	Customer slicing
DimProduct	ProductID, ProductName, Category, SubCategory, Brand	Text / Integer	Product slicing
DimEmployee	EmployeeID, Department,	Text / Integer	Employee analysis

	Region		
DimStore	StoreID, State	Text / Integer	Store analysis
FactSales	Sales, Cost, Quantity, Order/Delivery fields and keys	Decimal / Integer / Date / Text (validate exact model types)	Measures and transaction analysis
FactReturns	Return-related keys and quantities	Integer / Text / Date (validate exact model types)	Returns analysis
RegionTarget	TargetSales, TargetProfit, Region	Decimal / Text	Target benchmarking
SalesTargets	Target fields	Decimal / Text / Date as applicable	Target analysis; validate columns
ViewSelector	Selector field(s)	Text / Integer	UX / selection logic

The report uses fields including Customer ID/Name, Product ID/Name, Store State/ID, Total Sales and Total Quantity in the Executive Table View.

4. Row-Level Design

Recommended grain documentation: FactSales should be documented as the lowest business transaction grain available in the PBIX. Do not describe it as 'one row per order' unless the physical key confirms that. If an OrderID + ProductID combination is the grain, state that explicitly in the final technical specification.

Object	Required grain / key documentation	Current status
FactSales	Primary/business key and exact grain	Validate in Power BI Model/Data view
FactReturns	Return key and relationship key to sales/order	Validate
DimCustomer	One row per CustomerID	Conceptual expectation; validate duplicates
DimProduct	One row per ProductID	Conceptual expectation; validate duplicates
DimStore	One row per StoreID	Conceptual expectation; validate duplicates
DimEmployee	One row per EmployeeID	Conceptual expectation; validate duplicates
DimDate	One row per calendar date	Conceptual expectation

5. Schema Strategy – Star Schema

The preferred semantic-model pattern is a star schema: dimensions filter facts, while measures are calculated from fact tables. Keep filter direction from dimension → fact wherever possible. Avoid unnecessary many-to-many and bidirectional relationships.

Conceptual ER / Star-Schema View

Solid lines: model/report-supported usage. Exact physical cardinality and keys should be validated in Model view.

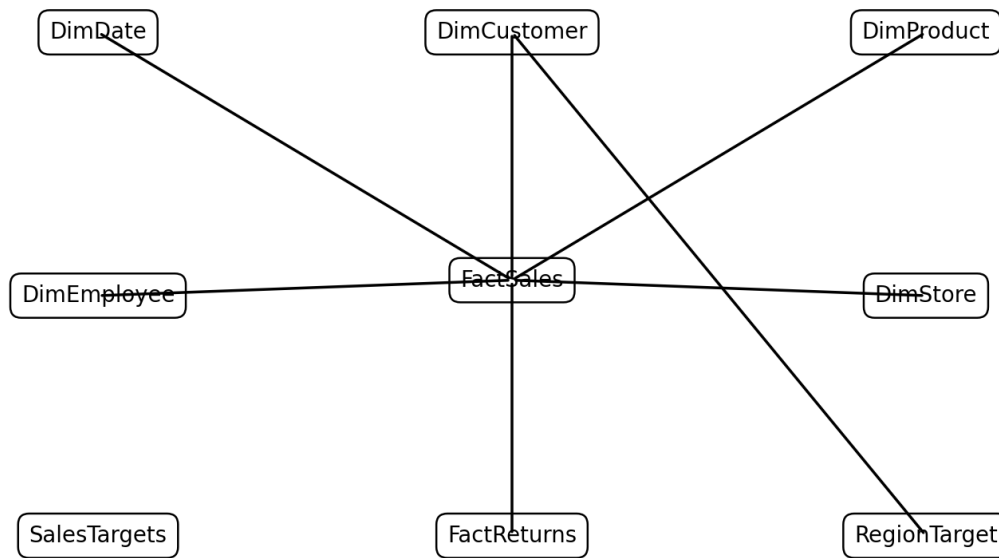


Figure 1. Conceptual ER / star-schema view. Exact physical keys/cardinality must be validated in Power BI Model view.

6. Relationship / Connection Matrix

From	To	Conceptual relationship	Recommended direction	Validation
DimDate	FactSales	Date key → transaction date key	Single: DimDate → FactSales	Confirm active date key
DimCustomer	FactSales	CustomerID → CustomerID	Single: DimCustomer → FactSales	Confirm 1:*
DimProduct	FactSales	ProductID → ProductID	Single: DimProduct → FactSales	Confirm 1:*
DimEmployee	FactSales	EmployeeID → EmployeeID	Single: DimEmployee → FactSales	Confirm 1:*
DimStore	FactSales	StoreID → StoreID	Single: DimStore → FactSales	Confirm 1:*
FactSales	FactReturns	Business return/order key	Prefer shared dimension/bridge where possible	Confirm actual design
DimCustomer	RegionTarget	Region → Region	Single if physically related	Confirm; may be better as shared Region dimension

The supplied report contains visuals that use FactSales together with DimDate, DimProduct and DimCustomer, and the Sales vs Target page uses DimCustomer.Region, FactSales measures and RegionTarget target columns.

7. How to Connect the Model in Power BI

1. Power BI Desktop → Home → Get Data → select the real source (Excel, SQL Server, CSV, etc.).
2. Load or transform source data in Power Query. Standardize column names, nulls, dates, data types and keys.
3. Create dimension tables first: DimDate, DimCustomer, DimProduct, DimEmployee and DimStore.

4. Load FactSales and FactReturns at their documented grain.
5. Create target tables such as RegionTarget/SalesTargets separately.
6. Open Model view and create key relationships. Prefer 1:* with the dimension on the 1-side and fact on the *-side.
7. Set cross-filter direction to Single unless a documented business requirement needs otherwise.
8. Hide technical key columns from report users after relationships are tested.
9. Create a dedicated Measures table if desired, then add DAX measures rather than calculated columns for aggregations.
10. Validate totals against source data before publishing.

For the current supplied report, the guide says the published report is opened from the Power BI Service workspace and that the dataset behind the report can be refreshed manually or by scheduled refresh.

8. KPI / DAX Reference

Measure	Business definition	DAX pattern
Total Sales	Sum of sales value	SUM(FactSales[Sales])
Total Profit	Sales minus cost	[Total Sales] - [Total Cost]
Total Cost	Sum of cost	SUM(FactSales[Cost])
Total Quantity	Sum of quantity	SUM(FactSales[Quantity])
Total Orders	Distinct count of orders	DISTINCTCOUNT(FactSales[OrderID])
Total Customers	Distinct customers	DISTINCTCOUNT(FactSales[CustomerID])
Profit Margin %	Profit / Sales	DIVIDE([Total Profit],[Total Sales])
Return Rate %	Returned orders / total orders	DIVIDE([Returned Orders],[Total Orders])
Avg. Order Value	Sales / orders	DIVIDE([Total Sales],[Total Orders])
Sales Growth %	Current vs prior comparable period	DIVIDE([Total Sales]-[Prior Sales],[Prior Sales])
YTD Sales	Year-to-date sales	TOTALYTD([Total Sales],DimDate[Date])
Late Delivery %	Late deliveries / relevant orders	DIVIDE([Late Deliveries],[Total Orders])
Avg. Delivery Days	Average order-to-delivery duration	AVERAGE(FactSales[DeliveryDays])

The business definitions for Total Sales, Total Profit, Profit Margin %, Total Orders, Return Rate %, Avg. Order Value, Sales Growth %, VIP customers, Late Delivery % and Avg. Delivery Days are documented in the supplied guide.

Important: the exact physical column names such as Sales, Cost, Quantity, OrderID, Date and DeliveryDays must be checked in the PBIX before passing these formulas. The formulas above are implementation patterns, not a claim that every physical column name is exactly this.

9. DAX Strategy Document

DAX layer	Strategy
Layer 1 – Base measures	Create reusable atomic measures first: Sales, Cost, Quantity, Orders, Customers, Returns.
Layer 2 – Profitability	Build Profit and Profit Margin from base measures instead of repeating SUM logic.
Layer 3 – Time intelligence	Use a dedicated marked Date table; build YTD, prior-period and growth measures from the date dimension.
Layer 4 – Ratios	Use DIVIDE() for percentage metrics to avoid divide-by-zero errors.
Layer 5 – Ranking	Use RANKX for Top N products/customers when required.
Layer 6 – Targets	Keep actuals and target measures separate; calculate variance and achievement percentage from them.
Layer 7 – Filter-aware metrics	Use CALCULATE() to intentionally modify filter context; avoid unnecessary ALL() because it can remove slicer context.

Layer 8 – Presentation	Format currency, counts and percentages at the measure level; keep report visuals thin and measure-driven.
------------------------	--

10. Target / Variance DAX Patterns

Target Sales = SUM(RegionTarget[TargetSales])

Target Profit = SUM(RegionTarget[TargetProfit])

Sales Variance = [Total Sales] - [Target Sales]

Sales Achievement % = DIVIDE([Total Sales], [Target Sales])

Profit Variance = [Total Profit] - [Target Profit]

Profit Achievement % = DIVIDE([Total Profit], [Target Profit])

11. Report / Visual Reference

Page	Purpose	Key content
Executive Dashboard	Management overview	KPI cards, profit trends, regional/product analysis and slicers
Sales vs Target	Benchmarking	Region-wise actual vs target table, combo chart and map
Executive Table View	Row-level detail	Customer, product, store and sales detail

The user guide documents three report pages and describes their purposes and visuals.

12. Power BI Service / Deployment Checklist

- ☐ Publish PBIX to the correct workspace.
- ☐ Confirm semantic model refresh succeeds.
- ☐ Configure gateway/credentials if the source requires an on-premises gateway.
- ☐ Set scheduled refresh only after manual refresh is successful.
- ☐ Validate RLS if security rules are implemented.
- ☐ Publish the report through an App when business users should consume a controlled navigation experience.
- ☐ Test report filters, totals and exports in the Service.

The supplied guide describes Share, Create app, navigation/audience, publishing, Refresh now and Scheduled refresh workflows.

13. References & Validation Notes

Primary reference: supplied Power BI User Guide – “TATA CLIQ Analytics Dashboard | Power BI User Guide”. The guide documents the model tables, report pages, KPI definitions, filtering behavior and Power BI Service workflow.

PBIX validation note: the supplied PBIX contains a DataModel artifact and a DiagramLayout containing the model objects. The extracted report metadata confirms the table names and many field/measure references, but the original physical source and exact column data types/cardinality were not reliably exposed by the available extraction path. Those items are therefore marked “validate” rather than invented.

14. Final Architecture Summary

Recommended architecture: Source → Power Query → Star Schema → DAX Semantic Layer → Power BI Report → Power BI Service. Use FactSales/FactReturns for transactional facts, reusable dimensions for filtering, target tables for benchmarking, and a layered measure strategy for maintainability and performance.